

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	58178	52.056192
1	[Git, Change, Create]	17320	15.497495
2	[Git, Commit]	11223	10.042054
3	[Git, Change, Delete]	8732	7.813171
4	[File, Git]	5885	5.265748
5	[Git, Change, Rename]	3344	2.992126
6	[Git, Tag]	1707	1.527380
7	[Author, Git, Person]	1263	1.130100
8	[Json, Key]	668	0.597709
9	[Json, Value, Scalar]	603	0.539549
10	[Committer, Git, Person]	370	0.331067
11	[NPM, Dependency]	338	0.302434
12	[Type, TS, Primitive]	291	0.260379
13	[Type, TS, Declared]	277	0.247853
14	[TS, ExternalDeclaration]	215	0.192377
15	[Git, Change, Copy]	138	0.123479
16	[Type, TS, Literal]	136	0.121689
17	[Json, Value, Object]	133	0.119005
18	[Type, TS, Union]	119	0.106478
19	[Type, TS, ObjectMember]	102	0.091267
20	[NPM, Script]	91	0.081424
21	[TS, Property]	65	0.058160
22	[TS, Function]	47	0.042054
23	[Git, Branch]	46	0.041160
24	[Type, TS, FunctionParameter]	40	0.035791
25	[Type, Object, TS]	39	0.034896
26	[File, Directory]	34	0.030422
27	[Type, TS, Function]	34	0.030422
28	[TS, Parameter]	33	0.029528
29	[Package, File, Json, NPM]	29	0.025948

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

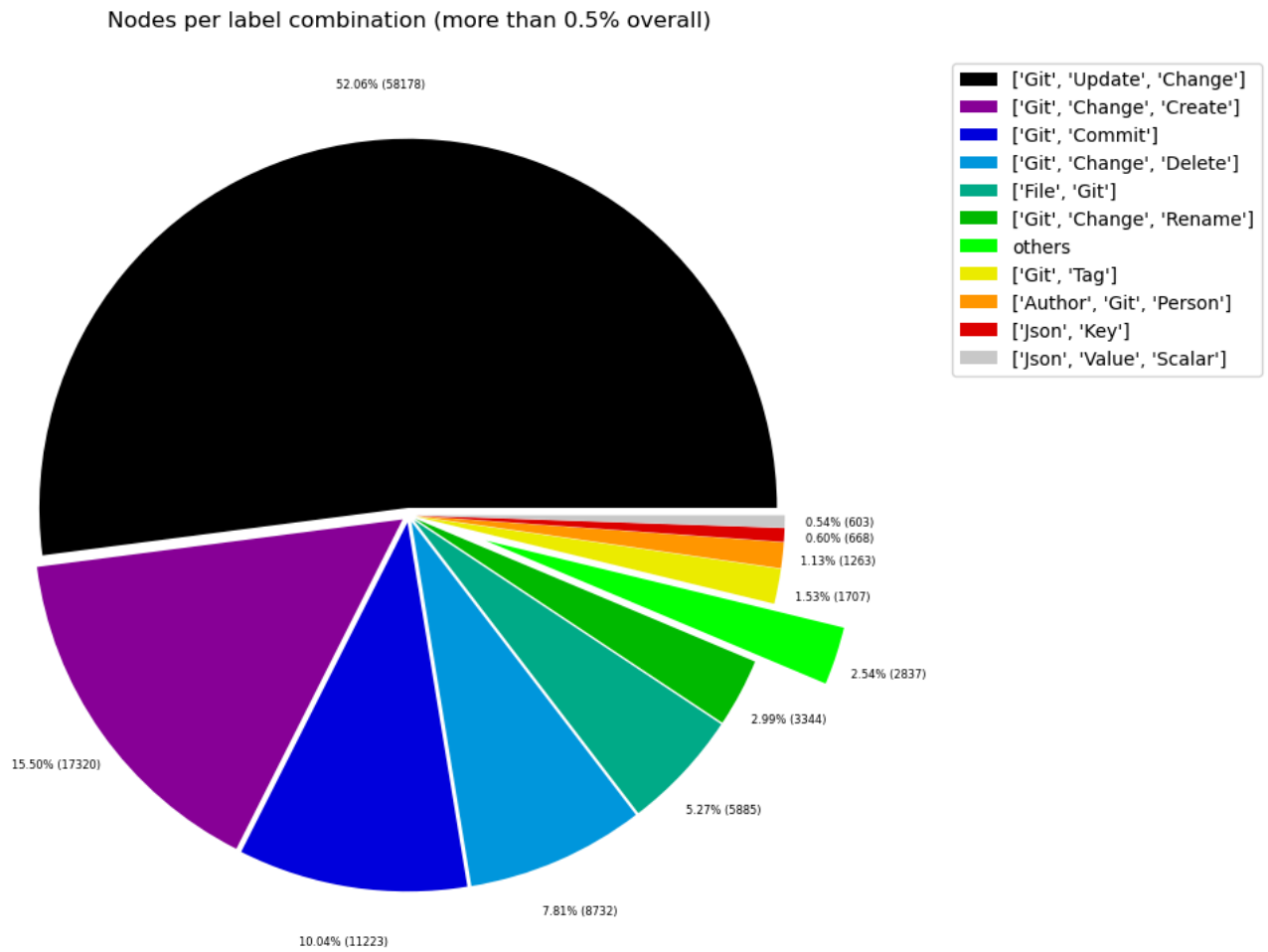


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.000895
1	[NPM, PublishConfig]	1	0.000895
2	[File, TS, Scan]	1	0.000895
3	[TS, Method]	1	0.000895
4	[Repository, File, Git]	1	0.000895
5	[TS, Constructor]	1	0.000895
6	[Value, TS, ObjectMember]	1	0.000895
7	[TS, Class]	1	0.000895
8	[TS, Enum]	2	0.001790
9	[Value, Object, TS]	3	0.002684
10	[Type, TS, Tuple]	3	0.002684
11	[Value, TS, Function]	4	0.003579
12	[TS, TypeParameter]	4	0.003579
13	[Value, TS, Complex]	5	0.004474
14	[Repository, NPM]	5	0.004474
15	[NPM, Engine]	6	0.005369
16	[Project, TS]	6	0.005369
17	[File, Local]	6	0.005369
18	[Type, TS, TypeParameterReference]	6	0.005369
19	[Value, TS, Member]	6	0.005369
20	[File, TS, Local, Module]	6	0.005369
21	[Value, TS, Call]	6	0.005369
22	[TS, EnumMember]	8	0.007158
23	[Type, TS, NotIdentified]	11	0.009843
24	[TS, ExternalModule]	11	0.009843
25	[Json, Value, Array]	12	0.010737
26	[Value, TS, Declared]	13	0.011632
27	[TS, TypeAlias]	16	0.014316
28	[File, Directory, Local]	16	0.014316
29	[TS, Interface]	17	0.015211

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

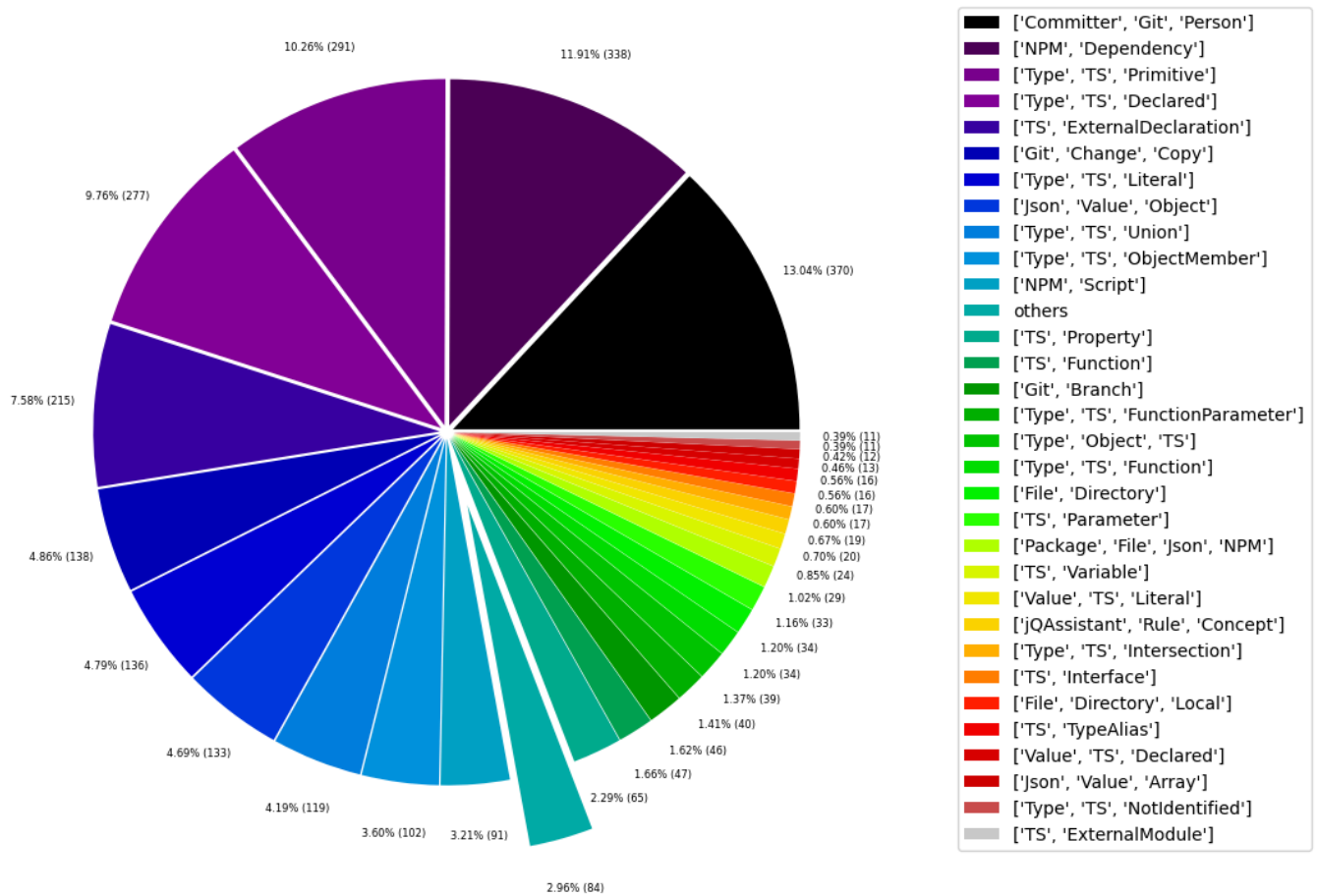


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

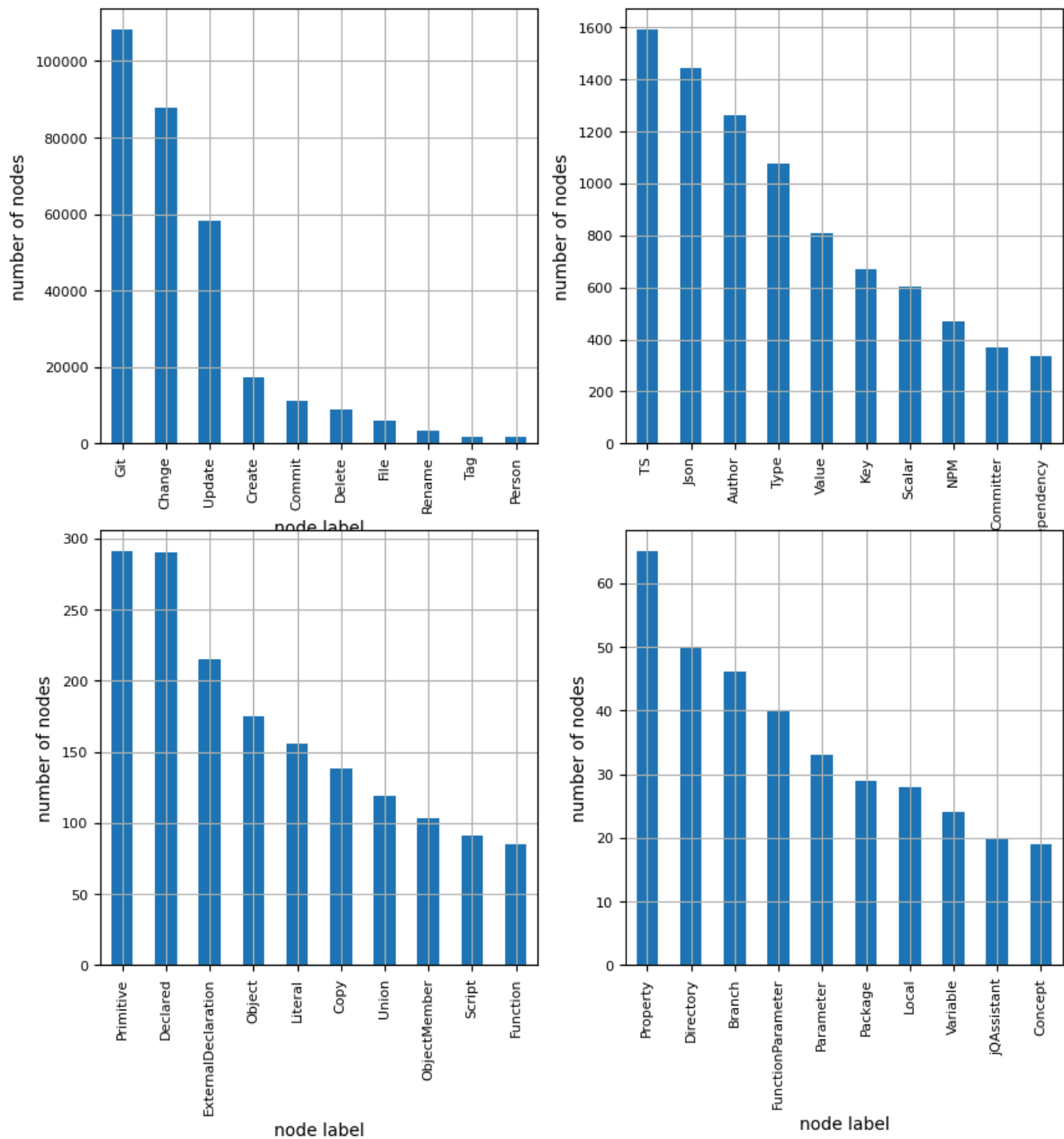
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	108207	96.820866
1	Change	87712	78.482462
2	Update	58178	52.056192
3	Create	17320	15.497495
4	Commit	11223	10.042054
5	Delete	8732	7.813171
6	File	5978	5.348962
7	Rename	3344	2.992126
8	Tag	1707	1.527380
9	Person	1633	1.461167
10	TS	1591	1.423586
11	Json	1445	1.292949
12	Author	1263	1.130100
13	Type	1075	0.961883
14	Value	806	0.721188
15	Key	668	0.597709
16	Scalar	603	0.539549
17	NPM	470	0.420544
18	Committer	370	0.331067
19	Dependency	338	0.302434
20	Primitive	291	0.260379
21	Declared	290	0.259485
22	ExternalDeclaration	215	0.192377
23	Object	175	0.156586
24	Literal	156	0.139585
25	Copy	138	0.123479
26	Union	119	0.106478
27	ObjectMember	103	0.092162
28	Script	91	0.081424
29	Function	85	0.076056
30	Property	65	0.058160
31	Directory	50	0.044739
32	Branch	46	0.041160
33	FunctionParameter	40	0.035791
34	Parameter	33	0.029528
35	Package	29	0.025948
36	Local	28	0.025054
37	Variable	24	0.021475
38	jqAssistant	20	0.017895
39	Concept	19	0.017001

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 333716

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	87712	26.283427
1	MODIFIES	87712	26.283427
2	UPDATES	58178	17.433386
3	COMMITTED	22446	6.726078
4	CREATES	20802	6.233444
5	HAS_PARENT	12303	3.686668
6	DELETES	12076	3.618646
7	HAS_COMMIT	11223	3.363039
8	HAS_FILE	5885	1.763476
9	RENAMES	3344	1.002050
10	HAS_NEW_NAME	1770	0.530391
11	HAS_TAG	1707	0.511513
12	ON_COMMIT	1707	0.511513
13	HAS_AUTHOR	1263	0.378466
14	DEPENDS_ON	887	0.265795
15	HAS_KEY	668	0.200170
16	HAS_VALUE	668	0.200170
17	CONTAINS	594	0.177996
18	HAS_COMMITTER	370	0.110873
19	OF_TYPE	338	0.101284
20	EXPORTS	309	0.092594
21	REFERENCES	197	0.059032
22	DECLARES	186	0.055736
23	DECLARES_DEV_DEPENDENCY	169	0.050642
24	DECLARES_DEPENDENCY	161	0.048245
25	COPIES	138	0.041353
26	HAS_MEMBER	103	0.030865
27	HAS_TYPE_ARGUMENT	94	0.028168
28	DECLARES_SCRIPT	91	0.027269
29	RETURNS	82	0.024572

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

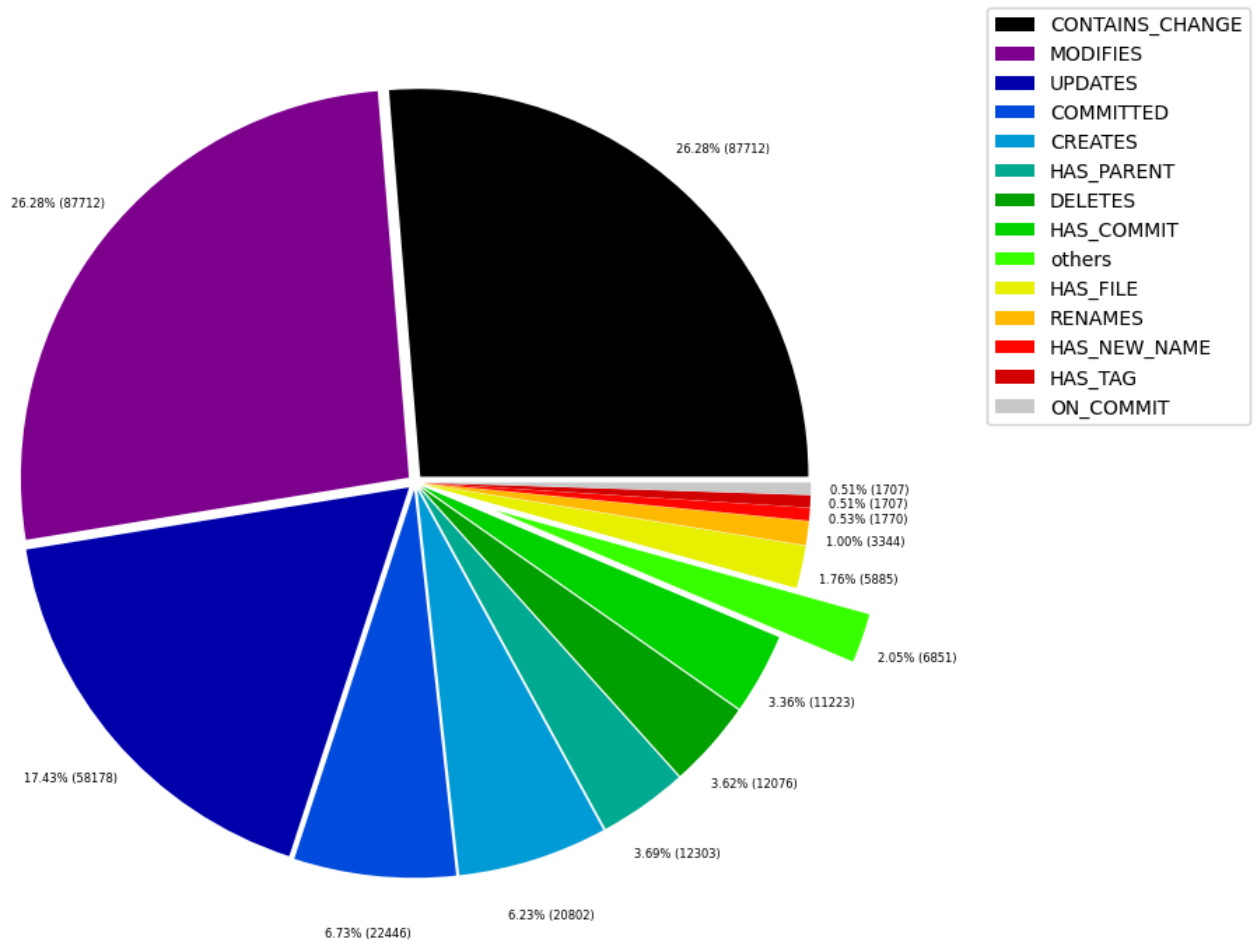


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

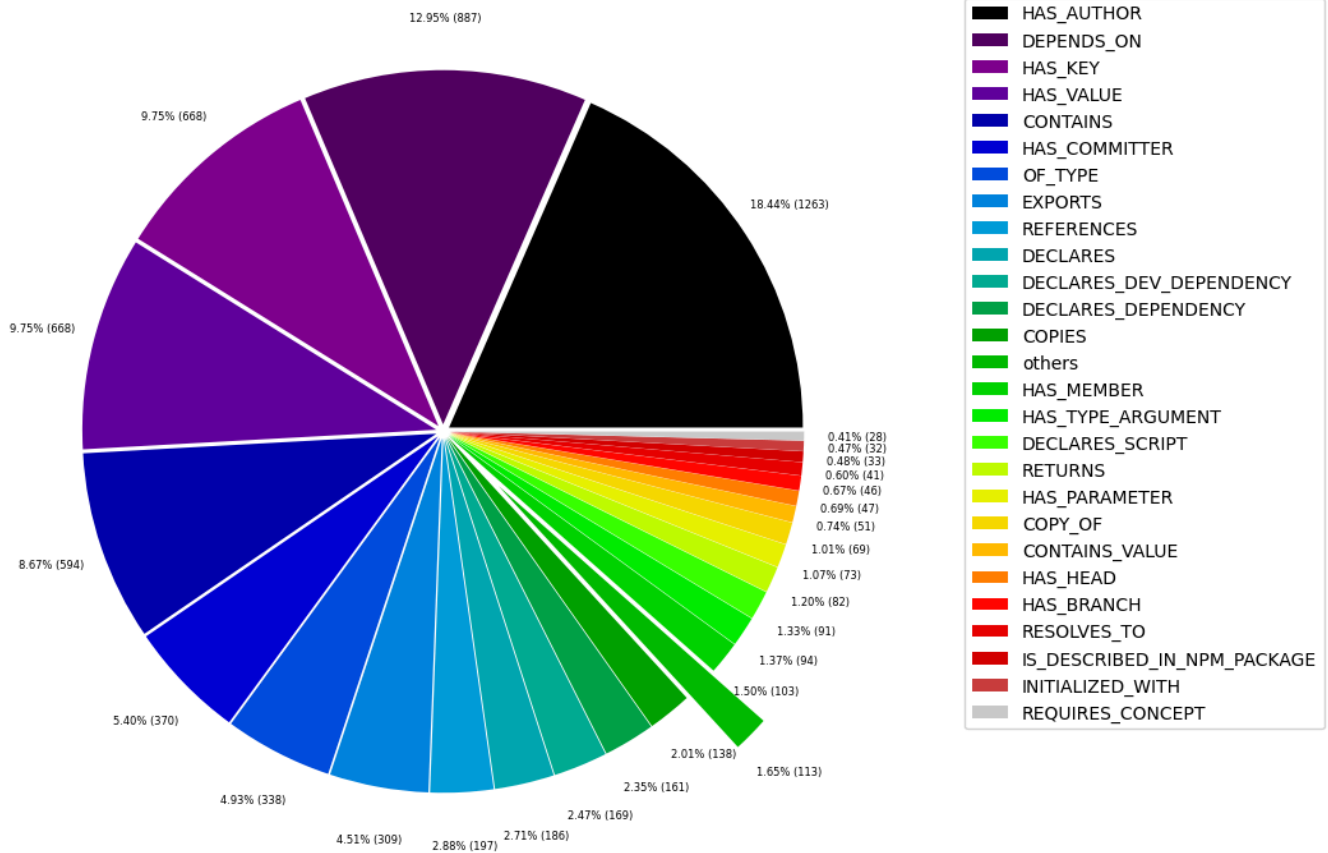
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	DECLARES_PUBLISH_CONFIG	1	0.000300
1	CONSTRAINED_BY	4	0.001199
2	IN_REPOSITORY	5	0.001498
3	REFERENCED_PROJECTS	5	0.001498
4	PARENT	6	0.001798
5	MEMBER	6	0.001798
6	HAS_ROOT	6	0.001798
7	HAS_NPM_PACKAGE	6	0.001798
8	HAS_CONFIG	6	0.001798
9	HAS_ARGUMENT	6	0.001798
10	EXTENDS	6	0.001798
11	DECLARES_ENGINE	6	0.001798
12	CONTAINS_PROJECT	6	0.001798
13	CALLS	6	0.001798
14	DECLARES_PEER_DEPENDENCY	8	0.002397
15	USES	11	0.003296
16	INCLUDES_CONCEPT	19	0.005693
17	REQUIRES_CONCEPT	28	0.008390
18	INITIALIZED_WITH	32	0.009589
19	IS_DESCRIBED_IN_NPM_PACKAGE	33	0.009889
20	RESOLVES_TO	41	0.012286
21	HAS_BRANCH	46	0.013784
22	HAS_HEAD	47	0.014084
23	CONTAINS_VALUE	51	0.015282
24	COPY_OF	69	0.020676
25	HAS_PARAMETER	73	0.021875
26	RETURNS	82	0.024572
27	DECLARES_SCRIPT	91	0.027269
28	HAS_TYPE_ARGUMENT	94	0.028168
29	HAS_MEMBER	103	0.030865

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOf
0	[Git, Update, Change]	MODIFIES	[File, Git]	58178	58178	
1	[Git, Update, Change]	UPDATES	[File, Git]	58178	58178	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	58178	11223	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	17320	11223	
4	[Git, Change, Create]	CREATES	[File, Git]	17320	17320	
5	[Git, Change, Create]	MODIFIES	[File, Git]	17320	17320	
6	[Git, Commit]	HAS_PARENT	[Git, Commit]	12303	11223	
7	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11223	1	
8	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11223	370	
9	[Author, Git, Person]	COMMITTED	[Git, Commit]	11223	1263	
10	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	8732	11223	
11	[Git, Change, Delete]	MODIFIES	[File, Git]	8732	8732	
12	[Git, Change, Delete]	DELETES	[File, Git]	8732	8732	
13	[Repository, File, Git]	HAS_FILE	[File, Git]	5885	1	
14	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	3344	11223	
15	[Git, Change, Rename]	CREATES	[File, Git]	3344	3344	
16	[Git, Change, Rename]	MODIFIES	[File, Git]	3344	3344	
17	[Git, Change, Rename]	RENAMES	[File, Git]	3344	3344	
18	[Git, Change, Rename]	DELETES	[File, Git]	3344	3344	
19	[File, Git]	HAS_NEW_NAME	[File, Git]	1770	5885	
20	[Repository, File, Git]	HAS_TAG	[Git, Tag]	1707	1	
21	[Git, Tag]	ON_COMMIT	[Git, Commit]	1707	1707	
22	[Repository, File, Git]	HAS_AUTHOR	[Author, Git, Person]	1263	1	
23	[Json, Value, Object]	HAS_KEY	[Json, Key]	668	133	
24	[Json, Key]	HAS_VALUE	[Json, Value, Scalar]	552	668	
25	[Repository, File, Git]	HAS_COMMITTER	[Committer, Git, Person]	370	1	
26	[TS, Function]	DEPENDS_ON	[TS, ExternalDeclaration]	293	47	
27	[File, TS, Local, Module]	DEPENDS_ON	[TS, ExternalDeclaration]	236	6	
28	[TS, ExternalModule]	EXPORTS	[TS, ExternalDeclaration]	215	11	
29	[Package, File, Json, NPM]	DECLARES_DEV_DEPENDENCY	[NPM, Dependency]	169	29	

Graph Density

```

total_number_of_nodes (vertices): 111760
total_number_of_relationships (edges): 333716
-> total directed graph density: 2.671825738022806e-05
-> total directed graph density in percent: 0.0026718257380228057

```